Practice Questions Sheet 1

This tutorial introduces some **basic ways of describing and visualising data** in R. Please **attempt all questions** before your tutorial You should work through the questions in a new R script. For questions asking you to interpret a result, **aim for one or two clear sentences**. You should use the context of the pizza delivery data when explaining your answers.

Student name: _____

Student number: _____

Please complete this assignment before the tutorial on the week of 12th October

If you cannot remember what a certain function does or a certain function's argument

If you are unsure what a function does, remember that you can use R's help system:

```
# What does the mean function do?
?mean()
```

You can also use the Help pane in RStudio.

Pizza delivery data

We will use a dataset containing information about **1,301 pizza deliveries**.
The dataset contains **four variables**:

Table 1: Overview of the pizza dataset

Variable	Description
Day	Day of the week on which the delivery was made
Distance	Distance between the pizza restaurant and delivery location, in km
Order amount	Value of the pizza order
Tip	Tip left by the customer

The dataset is available on the RStudio server in the ST201 data folder.
You can load it by opening a new R script and running the following code:

```
# Load the pizza dataset
pizza <- readRDS("Sharedfiles/ST201/data/pizza_del.rds")
```

Once you have loaded the data, you can look at the first few observations using:

```
# Examine the first 6 rows of the pizza dataset
head(pizza)
```

Exercise 1

- How many **observations** are in the dataset?
- How many **variables** are in the dataset?
- Which variables are **categorical** and which are **numerical**?

Individual observations

Let's start by looking at the `distance` variable.
Copy and run the below code

```
pizza$distance[1:10]
```

This displays the first 10 observations of distance.

> Remember that R uses square brackets `[]` to select observations.
>
> For example
>
> ```
> # The first observation
> pizza$distance[1]
> ```
>
> selects the **first observation**, while
>
> ```
> # The tenth observation
> pizza$distance[10]
> ```
>
> selects the **tenth observation**.

Exercise 2

- What is the value of the **275th observation** of `distance`?
- What does this **value represent** in the context of the dataset?
- What is the distance for the **1,000th observation**?

Measuring the centre of the data

Suppose the owner of the pizza restaurant wants to know the typical delivery distance.
One way to describe the centre of a numerical variable is using the mean.
In Figure 1, we can see the full distribution of the pizza delivery distance variable.



Figure 1: Histogram showing the distribution of pizza delivery distance (km)

To start off, copy and run the below code

```
mean(pizza$distance)
```

Exercise 3

- What is the **mean** delivery distance?
- Modify the above code and calculate the **median** delivery distance?
- Write one or two short sentences comparing the mean and median and then in the context of the data, compare the two values (are they similar, or noticeably different)

Quantiles

The mean and median tell us something about the centre of the data.
We can also use **quantiles** to describe where observations fall within the distribution.
Run the below code

```
quantile(pizza$ditance, )
```

Aha!! You should get an warning

Exercise 4

- Modify the above code to correctly calculate the 25th percentile
- Following this, calculate the 75th percentile
- Provide a short interpretation of these two quantiles, in the context of the pizza deliveries

Measuring variance

So far, we've been exploring the centre of `distance`. Let's zoom out a little and look at the spread of our variable

Exercise 5

- Calculate the **variance** of the `distance` variable
- Now, calculate the **standard deviation** of the `distance` variable
- In general terms, what is the difference between the term *variance* and the term *standard deviation*
- What is the **range** of the `distance` variable

Visualising data

Figure 2 is a box plot of the pizza `tips` variable

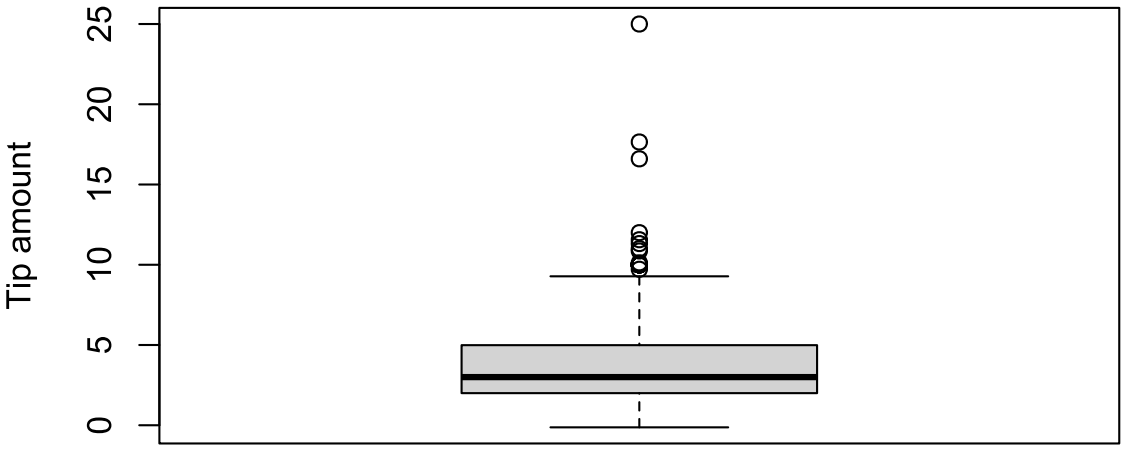


Figure 2: Boxplot of pizza tips

Exercise 6

- Write `R` code to produce the above plot (remember to add in the **y label**)
- Provide a short overview of what the boxplot is telling us (what is the median of the `tip` variable, what is the range, are there any outliers, et.)
- Modify the `R` code you just wrote to produce a histogram of the `tip` variable

Comparing delivery days

So far, we have been examining the data across all deliveries.
However, we can actually split this data up a little and look at each separate day (with the use of the `day` variable).
Suppose the restaurant owner wants to know whether customers spend different amounts on different days of the week.
Run the below code

```
# Create a dataset only with pizza deliveries on a Saturday
sat_pizza <- pizza[pizza$day == "Saturday", ]

head(sat_del)
```

This creates a new dataset containing Saturday deliveries only.

Exercise 7

- Using the new dataset, calculate the mean `order_amount` for Saturday deliveries.
- Create a dataset containing **Monday** deliveries only.
- Calculate the mean `order_amount` for Monday deliveries.
- Which day has the larger mean order amount?

i Bonus question

Creating a bunch of new datasets with different delivery days can be quite inefficient and tedious.
Imagine doing it 7 times and running seven different analyses!
Instead, a neat thing we can do is compare all the deliveries at once
Copy and paste the below code which outputs Figure 3

```
boxplot(tip ~ day, data = pizza, ylab = "Tip amount")
```

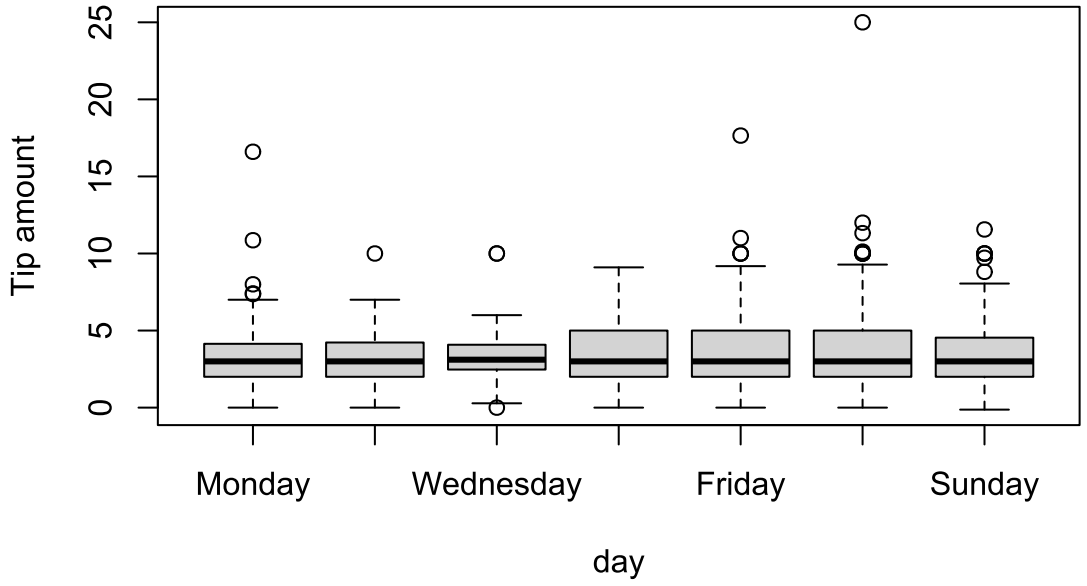


Figure 3: Overview of pizza tips across all delivery days

The ~ tells R to "*Plot tips separately for each value of day*".

- Which day appears to have the **lowest variation in tips**?
- Are there any potential outliers?